Project Assignment 2

Classifying Cell Images Using Pretrained CNN Models

Purpose:

This project is to exercise how to use deep CNN models to classify cell images. Efficient Human Epithelial-2 (HEp-2) cell image classification can facilitate the diagnosis of many autoimmune diseases.

Data:

The HEp-2 cell images are from the international competition on cell image classification hosted by International Conference on Pattern Recognition in 2014. The images have been pre-partitioned into training set (8701 images), validation set (2175 images), and test set (2720 images). In addition, a .csv file is enclosed. It contains the category of each image. This file consists of two columns: the first column is the image IDs of all the 13,596 images, and the IDs are consistent with the names of the images in the three sets; and the second column is the category of the cell image.

Tasks:

Please fine-tune two pretrained CNN models, AlexNet and ResNet34, for the six-class HEp-2 cell image classification task. Specifically, you need to:

(1) Initialize both models with ImageNet-pretrained weights and replace their final output layers with six-class classifiers.

(2) For each model, compare two strategies: partial fine-tuning (freeze early layers and train selected deeper layers plus the output layer) and full fine-tuning (train all layers). For example, partial fine-tuning may train FC6–FC8 in AlexNet and layer4 plus fc in ResNet34.

(3) Train and validate all four configurations using the provided data splits; use the validation set to select hyperparameters.

(4) Evaluate the selected models on the test set. Compare accuracy, trainable parameter counts and training time, and discuss the effects of architecture and fine-tuning strategy.

Hint: -Use the validation set to select the hyperparameters, e.g., the number of epochs in the training stage.

-Use data augmentation can help improve the classification performance. You can follow the data augmentation in the attached cell image classification paper.

Submission:

Please submit (1) **a complete report** (more instructions could be as follows)

(2) **the code used in your experiment** (in .zip/.rar/.7z format, please make sure you have your requirement packages included if you used other packages, **data are not included**)

As for the report, please submit individual reports and clearly indicate the contribution of each team member at the beginning of the report. It is suggested that the report is organized in the following ways. **The report should be in .pdf or .doc/.docx format.**

Section 1: Aims and Background

Section 2: Contributions of Team Members

Section 3: Methodology

Section 4: Results

Section 5: Discussion and Conclusion

Submission Deadline:

This project is due on**: 11:59pm, Nov 8, 2026 (Week 13 Friday)**

This project occupies **15% of the total marks**.

Evaluation Criteria:

Criteria	Exceptional	Good	Average	Poor
Introduction: Aims (5%)	5.0 pts — Exceptional Defines and elucidates the problem. Develops compelling rationale for the lab. Persuasively explains the multifactorial contribution of learning gained from the lab.	3.0 pts — Good Defines problem with some depth. States rationale for the lab. Provides explanation beyond basic contribution.	2.0 pts — Average Defines problem. States only obvious rationale for the lab. Explains the basic contribution of the activities to the understanding of the lab.	1.0 pts — Poor Shows fundamental lack of understanding. Does not state rationale for the lab. Does not explain the contribution of the activities to the understanding of the lab.
Background and Methods: Knowledge of the area and use of sources (25%)	25.0 pts — Exceptional Demonstrates exceptional depth of knowledge of the field. Comprehensive use of most recent and seminal sources. Clearly discriminates among seminal sources.	20.0 pts — Good Demonstrates proficient knowledge of the field. Thorough selection of sources pertinent to project. Shows some discrimination among seminal sources.	15.0 pts — Average Demonstrates a basic knowledge of the field. Selected sources relevant to project. Limited discrimination among seminal sources.	7.0 pts — Poor Lacks a basic knowledge of the field. Selected sources irrelevant to project. Misinterprets sources.
Results (50%)	50.0 pts — Exceptional Results organised into sections and subsections in a logical order.	45.0 pts — Good Results are well organised in a logical order.	35.0 pts — Average Results organised into sections and subsections.	10.0 pts — Poor Results are poorly organised.
Discussion and Conclusion (20%)	20.0 pts — Exceptional Outstanding and innovative synthesis between results and sources.	15.0 pts — Good Demonstrates synthesis between results and sources.	10.0 pts — Average Relevant information about the results are discussed under each heading. Demonstrates information synthesis to a limited extent.	5.0 pts — Poor Information about the results are repeatedly reported without any synthesis from sources.
Total Points: 100.0				